Introduction to Pandas

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`Pandas` is the central element of the `Python` ecosystem for *data science*. The recent success of `Python` in data analysis owes a lot to `Pandas`, which made it possible to import `SQL` logic into the `Python` language. `Pandas` embeds a huge number of features that allow efficient processing pipelines for medium-sized data (up to a few gigabytes). Beyond this volume, other solutions will be needed (`DuckDB`, `Dask`, `Polars`, `Spark`...).

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The `Pandas` part has recently evolved. You can find the content related to `Pandas` in the following chapters:

- Introduction to `Pandas` and the logic of the *DataFrame*;
- Combining data with `Pandas`: joins;
- Handling data with `Pandas`: statistics and reshaping;
- Pandas: limits and alternative ecosystems;

Informations additionnelles

ℹ Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
```

```
    "folium>=0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables>=0.12.0",
    "gt-extras>=0.0.8",
    "ipykernel>=6.29.5",
    "jupyter>=1.1.1",
    "jupyter-cache>=1.0.0",
    "kaleido>=0.2.1",
    "langchain-community>=0.3.27",
    "loguru==0.7.3",
    "markdown>=3.8",
    "nbclient>=0.10.0",
    "nbformat>=5.10.4",
    "nltk>=3.9.1",
    "openpyxl>=3.1.5",
    "pandas>=3.0",
    "pip>=25.1.1",
    "plotly>=6.1.2",
    "plotnine>=0.15",
    "polars>=1.8.2",
    "py7zr>=1.1.3",
    "pyarrow>=17.0.0",
    "pynsee>=0.1.8",
    "python-dotenv>=1.0.1",
    "python-frontmatter>=1.1.0",
    "pywaffle>=1.1.1",
    "requests>=2.32.3",
    "scikit-image>=0.24.0",
    "scikit-learn>=1.8.0",
    "scipy>=1.13.0",
    "seaborn>=0.13.2",
    "selenium<4.39.0",
    "spacy>=3.8.4",
    "webdriver-manager>=4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]

[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean>=4.0.1",
```

```
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.

Bibliography